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STAFF REPORT FOR THE 2014 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITYANALYSIS

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The Debt Sustainability Analysis (DSA) has been prepared jointly by IMF and World Bank staff, in consultation with the authorities, using the debt sustainability framework for low-income countries approved by the Boards of both institutions.

This debt sustainability analysis (DSA) concludes that the Republic of Congo faces a low risk of debt distress, similar to the analysis prepared in the previous year.[8] Under the baseline scenario for the current DSA all debt burden indicators are projected to remain below the policy-dependent indicative thresholds. Standard stress tests result in one breach of these thresholds which could give rise to a downgrade of Congo's rating to "moderate risk of debt distress". This finding underscores the vulnerability of the Congolese economy to external shocks, in particular, to adverse changes in oil prices and external demand. However, in view of the Republic of Congo's large government deposits at the BEAC and gross official reserves at the BEAC, which gradually increase in the medium-term under the baseline scenario, and which are a mitigating factor that could help Congo weather potential adverse shocks, Congo is classified as at "low risk of debt distress". In order to increase the resilience of the Congolese economy, efforts should continue to focus on diversification of the tradable sectors. Moreover, medium-and long-term fiscal and debt sustainability can be promoted through continued adherence to prudent borrowing policies and building of fiscal buffers. Under an alternative policy scenario, which assumes a continuation of large government spending over the medium term, gross official reserves would quickly deteriorate and debt-burden indicators would be more unfavorable. Under the alternative policy scenario debt vulnerabilities would be significantly higher and entail a rating downgrade."

[8] This DSA is based on end-2013 debt data. The 2013 DSA (see IMF Country Report No. 13/282) was based on end-2012 debt data. The fiscal year for the Republic of Congo is January–December.

BACKGROUND

1. Congo's stock of external debt has been building up since the extensive HIPC/MRDI debt relief in 2010. As noted in the previous DSA, Congo reached the HIPC Initiative Completion Point in January 2010. This resulted in estimated total debt service savings of US$1.9 billion, and a decline in gross public external debt to just over 20 percent of GDP at end-2010 (from about 55 percent of GDP in 2009) (See Press Release No. 10/20, January 28, 2010). However, the external debt stock at the end of 2013 amounted to around US$4.3 billion, up from US$2.4 billion in 2010.[9] The debt-to-GDP ratio increased steadily over the past 3 years, reflecting new borrowing, notably large disbursements under bilateral loan agreements with China, contracted in 2006 and 2012. Bilateral creditors currently comprise nearly two-thirds of total debt (see Table 1). Domestic public debt is relatively low at around 4 percent of GDP. The bulk of domestic arrears involve pension arrears and unpaid social benefits which started accumulating after the liquidation of public enterprises in the past as employees were not rightfully terminated.[10]

Table 1. Republic of Congo: Gross External Debt by Creditor and Total Debt service

	2010		2011		2012		2013	
	CFA billion	USD million	CFA billion	USD million	CFA billion	USD million	CFA billion	USD million
Debt stock	1,203	2,434	1,568	3,325	1,792	3,511	2,132	4,316
Multilateral	227	459	90	191	97	190	95	193
Bilateral	291	589	957	2,031	1,187	2,325	1,538	3,113
of which China	182	369	761	1,615	1,014	1,987	1,401	2,837
Private Creditors	685	1,386	520	1,104	508	996	499	1,010
Total debt service	72.6	146.8	65.7	139.3	69.9	137.1	150.7	305.1
Interest	15.7	31.8	13.4	28.5	11.9	23.3	16.9	34.1
Amortization	56.9	115.0	52.2	110.8	58.1	113.8	133.8	270.9
Multilateral	8.2	16.5	7.4	15.6	7.0	13.8	10.4	21.0
Bilateral	23.6	47.7	37.5	79.5	47.2	92.5	114.8	232.4
Private Creditors	25.1	50.8	7.4	15.6	3.8	7.5	8.7	17.6

Source: Caisse Congolaise d'Amortissement (CCA) and IMF staff calculations

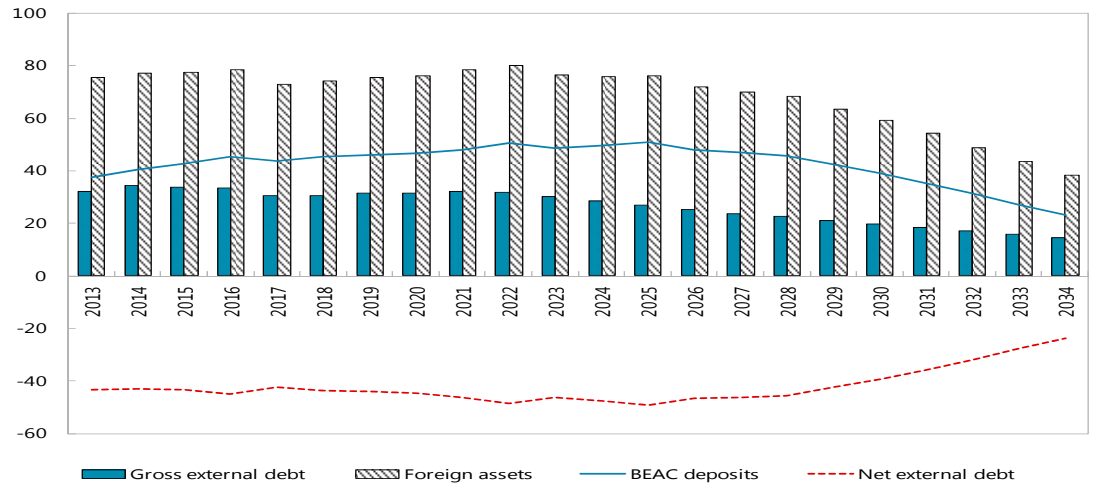
2. The sustainability of Congo's debt should be assessed in the context of the country's sizeable financial assets. Gross official reserves stood at about 38 percent of GDP at end-2013. In addition, the government holds substantial foreign currency deposits at China's EXIM bank. As depicted in Figure 1, the government's large deposits at the BEAC are equivalent to about

[9] This figure includes also the external arrears stock of US$739 million. More than ⅔ are related to commercial creditors, for which legal decisions are pending. The rest is owed to bilateral creditors for which the provisions of debt relief under the HIPC Initiative are still under discussion.

[10] The authorities are currently undertaking an audit of domestic arrears. Once the audit is completed, the authorities intend to put in place a multi-year repayment plan.

60 percent of external public debt and gross official reserves at the BEAC are equivalent to about 118 percent of external debt. The large gross official reserves and government deposits at the BEAC reflect the ongoing run of high international oil prices since the achievement of HIPC/MDRI debt relief in 2010. Assuming the government continues to save part of its annual oil revenues, Congo's gross official reserves will continue to exceed the level of public debt throughout the projection horizon.[11]

Figure 1. Republic of Congo: Gross and Net External Position



Sources: Congolese authorities; IMF and WB staff calculations and projections.

3. Nonetheless, there is a need to strengthen public debt management. A recent IMF public debt management technical assistance needs assessment mission highlighted the need to: (i) strengthen the legal and institutional framework to cover the entire process of issuing and managing debt; (ii) develop a medium-term debt management strategy and enhance transparency in debt management; (iii) introduce a formal framework for managing the government's cash flows; and (iv) strengthen the staff analytical capabilities with respect to managing the public debt.

UNDERLYING ASSUMPTIONS

4. The medium and long-term macroeconomic framework underlying the DSA is consistent with the baseline scenario presented in the Staff Report for 2014 Article IV Consultation. The main assumptions and projections for key macroeconomic variables are summarized in Box 1. Economic activity is expected to remain strong, largely driven by the non-oil

[11] The Congolese authorities are receiving ongoing technical assistance from the IMF's Monetary and Capital Markets Department to strengthen debt management.

sector, reflecting developments in construction, agriculture and services. Oil production is foreseen to gradually increase from 88 million barrels in 2013 to a peak of 118 million barrels in 2017 and then decline in later years. Large capital expenditures are projected to continue in the medium term to address the country's infrastructure gap. Once a critical amount of public capital stock is reached, corresponding to middle-income countries' level, public investment is assumed to gradually decline as private investment picks up, attracted by an improved infrastructure base and a friendlier business climate.

5. Assumptions regarding external borrowing have been adjusted compared to the macroeconomic framework underpinning last year's DSA (see Table 2). The projected debt profile in last year's DSA only reflected already committed loans. In this year's DSA, the debt profile is substantially higher as it also includes the authorities' medium-term borrowing plans. The Congolese authorities are projected to continue borrowing externally at broadly the same pace as in the last three years. In line with these projections, new gross external borrowing over the medium term (2014–19) is expected to total around US$3.3 billion. The terms of the new debt are assumed to be highly concessional, in line with the agreement signed with China in 2012.[12] The degree of concessionality is assumed to lessen after 2025. Compared to previous DSAs, the grant element is higher, mainly due to the assumed upward revision of the discount rate (from 3 to 5 percent).[13]

6. Under the baseline, the current account balance is expected to be negative over 2014–16, but it turns positive in later years. These developments are mainly driven by ambitious infrastructure projects and new investments in the oil sector that result in an upward trend in imports of goods and services. The *Moho Nord* deep offshore field, developed by Total jointly with Chevron and SNPC, is expected to attract around $10 billion of foreign direct investments over 2013–16, and to come on stream in 2017 when Congolese oil production is expected to peak. Considering the large uncertainty in the mining sector, a conservative approach is adopted; only one of Congo's iron ore projects under development is assumed to reach the production phase, containing output growth. Removing existing sector-specific infrastructure bottlenecks in transportation would provide a significant boost to longer-term growth at the cost of significant initial investments. In the long run, as oil exports decline and non-oil export receipts fail to compensate for this loss, additional financing would be needed. The present DSA assumes that the financing gap after 2025 is filled on concessional terms, although with a lower grant element than that under the current Chinese loans.[14]

[12] This assumption reflects the high level of deposit collateralization, which at this moment is estimated to be around 150 percent.

[13] The 2014 DSA's grant element of new public sector borrowing in the short and medium term is around 49 percent, reflecting 5 years of grace period, 20 years maturity and very low interest rate on loans.

[14] The grant element of new borrowing in the long term is assumed to be around 28 percent, mainly reflecting higher interest rates.

Table 2. Republic of Congo: Comparison of PPG External Debt, Baseline Scenario
(In percent of GDP, unless otherwise indicated)

	2014	2015	2016	2017	2018	2019	2024	2032
				PV of Debt to GDP Ratio				
2014DSA	27.0	25.4	24.3	21.2	20.6	20.6	19.0	13.7
2013DSA	18.3	16.5	14.9	12.0	10.5	9.2	7.5	9.0
				PV of Debt to Exports Ratio				
2014DSA	34.8	33.6	33.1	28.6	28.7	30.3	36.3	35.1
2013DSA	25.4	23.1	21.6	17.1	15.6	14.8	16.5	28.7
				PV of Debt to Government Revenue Ratio				
2014DSA	58.6	57.5	55.9	52.7	52.8	54.7	51.9	43.3
2013DSA	39.2	39.1	35.7	30.4	27.5	24.9	22.1	28.8
				Debt Service to Exports Ratio				
2014DSA	3.5	3.9	3.7	3.3	3.4	3.5	2.4	2.7
2013DSA	2.9	2.7	2.6	2.3	2.3	2.6	1.3	1.7
Memorandum Items								
				New Loan Disbursements (billions FCFA)				
2014DSA	374	300	280	260	240	220	108	139
2013DSA	125	75	50	1	1	1	202	139
				Grant Element of New External Borrowing (in percentage points)				
2014DSA	48.7	48.7	48.7	48.7	48.7	48.7	48.7	28.1
2013DSA	32.9	32.9	32.9	23.8	23.8	23.8	23.8	23.8
				Basic non-oil primary balance (percent of the non-oil GDP)				
2014DSA	-48.5	-44.8	-40.0	-35.3	-30.7	-26.5	-7.8	0.1
2013DSA	-39.4	-36.1	-32.2	-29.1	-26.6	-24.0	-9.3	-3.0
				Real GDP growth (annual percentage change)				
2014DSA	6.0	7.5	7.3	9.0	3.5	1.9	3.9	4.5
2013DSA	4.8	7.7	7.3	10.1	4.7	2.7	4.3	4.2
				Current Account Balance (percentage of the GDP)				
2014DSA	-3.1	-3.5	-1.4	2.3	1.4	-1.6	-2.2	-2.1
2013DSA	5.2	0.9	0.8	2.7	0.2	-3.2	-8.5	-11.9

Sources: Congolese authorities; IMF and WB staff calculations and projections.

Box 1. Republic of Congo: Main Macroeconomic Assumptions

- **Real GDP growth:** The non-oil sector is projected to grow on average by 7 percent per year during 2014–21 driven by construction, agriculture and public investment; non-oil growth would stabilize at about 5 percent thereafter. Given the uncertainties surrounding the mining projects (including sluggish global demand and the need for large investment in infrastructure) the baseline projections for mining output are cautious. Iron ore production, which is expected to start timidly in 2014 with 200,000 tons, is supposed to augment to 2 million tons in 2015, to 7 million tons by 2017, and to 17 million tons by 2023. These projections are conservative, since the estimated potential of mining in Congo is more than 50 million tons per year if infrastructure gaps are addressed.[15]
- **Oil production and prices:** Annual oil production is projected at around 89 million barrels in 2014 and expected to peak at 118 million barrels in 2017. Based on official projections, production would steadily decline over 2019–34, to about 13 million barrels in 2034. Following projections in the IMF's *World Economic Outlook*, international oil prices are projected to decrease by about 3 percent per year during 2014–18; thereafter, they are assumed to remain constant in real terms.
- **Inflation:** Following the sharp acceleration to 7.5 percent (year-on-year) in 2012, inflation decelerated to close to 2 percent at the end of 2013 in line with regional and international trends. The appreciation of the CFA franc with respect to the U.S. dollar also helped to anchor price stability. In the medium term, inflation is projected to stay below the CEMAC's convergence criteria of 3 percent.
- **Current account balance:** Current account fluctuations are driven by the dynamics of exports and imports in the oil and mining sectors. The balance is expected to be positive when oil production peaks in 2017 and 2018; however, it would turn to a widening deficit thereafter as the currently assumed mining production would not suffice to compensate for the decline in oil production.
- **Fiscal balance:** Following the budgeted sharp increase in public capital expenditures in 2014, the baseline projections assume that fiscal consolidation resumes in 2015 and continues in the medium- and long-term. The basic non-oil primary deficit would narrow from about 48.5 percent of non-oil GDP in 2014 to 26.5 percent in 2019 and close to zero in the long run.
- **External financing:** In the medium term, the authorities are assumed to continue to borrow externally at similarly concessional terms. The second loan agreement with China worth US$ 1 billion is assumed to be disbursed over the 2013–14 period. This is assumed to be followed by additional borrowing over the period 2015–21 to the tune of another US$ 3.3 billion. These disbursements would finance reconstruction efforts in the aftermath of the 2012 ammunitions depot explosions, contribute to address Congo's infrastructure gap and help the preparations of the 11th All-Africa Games organized in Brazzaville in 2015. In the long run, external borrowing would be needed to ensure a minimum public investment of about 9 percent of GDP. The concessionality level is expected to lessen after 2025.

[15] Fluctuations in real GDP growth from 2018 onward mainly reflect fluctuations in oil production with the exception of 2023 when mining production is expected to increase further.

7. The macroeconomic outlook is subject to risks. The baseline scenario is built around relatively favorable assumptions about commodity prices and the availability of new public sector borrowing on concessional terms. Given the high reliance on oil revenues and exports, the Congolese economy remains vulnerable to external shocks, notably to a fall in oil prices or a slowdown of trade partners' demand. However, the Republic of Congo has accumulated sufficient fiscal and external reserves to mitigate the immediate effects of such shocks. Nevertheless, a protracted worsening of the terms of trade or foreign demand could trigger severe imbalances and require additional borrowing to avoid a sharp contraction in income. This could have negative implications for debt sustainability.

8. To illustrate the vulnerabilities facing Congo, a DSA under an "alternative policy scenario" that assumes a continuation of current large spending is also prepared. The important policy decisions facing the authorities are highlighted by a comparison of the baseline scenario with the "alternative policy" scenario (see Table 8).

DEBT SUSTAINABILITY ANALYSIS

G. External DSA

9. Under the baseline, Congo's debt indicators are below the relevant country-specific debt burden thresholds. The joint Bank-Fund debt sustainability framework (DSF) for low-income countries classifies Congo as a "weak" performer, based on the quality of the country's policies and institutions as measured by the 3-year average of the ratings under the World Bank's Country Policy and Institutional Assessment (CPIA). This is reflected in lower debt sustainability thresholds compared to countries operating in a strong policy environment.[16] Nevertheless, given the low level of external debt and strengthening indicators of repayment capacity, the debt stock and debt service ratios remain comfortably within the sustainable debt domain throughout the projections period under the baseline.

10. One of the stress tests breaches temporarily the indicative threshold. Standard bound tests that examine the implications of various shocks to the debt and debt-service paths based on the country's historical volatility result in sharp increases in the debt burden indicators (Figure 2a). For all indicators, except for the debt-service to revenue ratio, the most extreme scenario is associated with a shock to exports. A decline in exports to a level equivalent to one standard deviation below their historical average in the first two years of the projection period would cause the PV of debt to GDP ratio to rise by about 14 percentage points and the PV of debt to export ratio by 35 percentage points at the peak.[17] This development results in a breach of the threshold in the case of the PV of debt to GDP path during 2016–23. The breach of the threshold relates to increased

[16] The thresholds for weak performers are 100, 30 and 200 percent for the PV of debt to exports, GDP and government revenue, respectively. Debt service thresholds are 15 and 18 percent of exports and revenue, respectively.

[17] The vulnerability stems from the very high share of oil in Congo's exports and the large volatility of oil prices and production. The magnitude of the shock doubled to -10 percent in this exercise compared with last year's DSA.

oil export volatility in recent years that results in a doubling of the magnitude of the export shock compared with the magnitude of the shock in last year's DSA. While this breach of the policy-dependent threshold would give rise to a downgrade of Congo to a "moderate risk of debt distress," we consider that the higher volatility of the oil exports is temporary and is expected to be compensated by a higher production in the medium term. In addition, Congo's large external asset position would help to buffer it from an adverse export shock. Under the historical scenario, which derives the debt indicators assuming that key variables are at their 10-year historical averages, all debt ratios decline very rapidly. However, as pointed out in the previous DSA, this scenario is less relevant for resource-rich countries since past trends are likely to be a poor predictor of future outcomes.

H. Public DSA

11. Adding domestic public debt to external debt does not change the results of the analysis (see Figure 3). Congo's domestic public and publicly guaranteed (PPG) debt includes domestic arrears. The evolution of the total public debt indicators broadly follows that of external debt under the baseline. The most extreme shock (in this case real GDP growth in 2014-15 set at one standard deviation below its historical average) raises the PV of debt to GDP ratio substantially to over 50 percent in the outer years of the projections period, and stabilizes at this level. A permanently lower GDP growth would have a similar, albeit somewhat smaller, effect on the debt ratios. The accumulation of official foreign reserves, which in the case of Congo is significant due to high primary surpluses, can be regarded as a buffer against these potential shocks.

I. External and Public DSA in an Alternative Policy Scenario

12. External and Public DSAs were also ran for the case in which an "alternative policy" scenario is used as the baseline. Under the alternative policy scenario, continued large government capital spending together with further real appreciation from higher inflation would lead the current account to register increasing deficits from 2015 onwards. As a result, reserves would fall to 5 months of imports by 2019—which is on the low side for a country with a highly concentrated export basket and a pegged exchange rate, and the government would eventually need to borrow more to finance its spending. As a result, the debt-to-GDP ratio is projected to exceed 50 percent by 2020 and to top 66 percent by 2025. Under the alternative policy scenario, new loans would likely be less concessional from 2020 onwards. Interest rates would rise to 5 percent, on average, and the average grace period would shorten. Eventually, the rise in external debt would force the inevitable downward adjustment in government spending, and growth would fall below the average growth rates in the "strong policy" scenario.

13. Under the alternative policy scenario, debt vulnerabilities would be significantly higher and may entail a rating downgrade." Figure 2b illustrates that the PV of the debt-to-GDP ratio would generally remain above the indicative threshold of 40 percent from 2019 onward. Moreover, the PV of debt-to-exports ratio would breach the threshold from 2021 onwards and the debt service-to-exports ratio would reach the threshold in 2030.

AUTHORITIES VIEW

14. The Republic of Congo's authorities broadly concurred with the overall assessment of low risk of debt distress, highlighting the strong net external position that would provide a buffer against adverse shocks. The authorities reiterated that they do not want to repeat the experience from the beginning of the 2000s, when debt exceeded by far the CEMAC convergence criteria. They assured staff of their plans to continue to borrow only under highly concessional terms. Finally, the authorities are aware of the risks related to oil prices and exports and their strategy is to continue the economic diversification.

Box 2. Macroeconomic Assumptions: Strong vs. Alternative Policy Scenario

1. Under the baseline "strong policy" scenario, the authorities undertake fiscal consolidation and build up their fiscal and external buffers in the medium to long term. Under this "strong policy" scenario, the government implements a fiscal consolidation, with government spending remaining broadly flat in nominal terms over the period 2014–19. Growth continues on average at a healthy clip in the medium to long-term, even in the face of a contraction in the oil sector. Inflation remains moderate. The current account is a deficit of one percent on average over 2014–19, slipping into an average deficit of 2 percent over the medium to long-term. Gross official reserves average 10 months of prospective imports of goods and services over 2014–19, and coverage is expected to remain above that ino the longer-term. The initial fiscal consolidation frees up fiscal space for higher average expenditure growth over 2027–34 (Table 8, "Baseline").

2. The alternative policy scenario illustrates the risks of continued large government spending. In this scenario, government spending increases unchecked to a peak of 105 percent of GDP in 2018. As a result, overall growth would on average be higher than in the strong policy scenario over the period 2014–19, yet already in the period 2020–26 growth would dip below that of the baseline. (Table 8, "Alternative Policy").

3. However, the short term boost to growth comes at a significant longer-term cost. Higher imports cause the current account to deteriorate significantly over the period 2020–26, with the deficit remaining on average wider than in the strong policy scenario. Higher government spending also pushes up inflation compared to the baseline scenario, eroding competitiveness. Gross official reserves fall below external public debt. A sharp correction in government spending is required to correct the imbalances.

4. The keys assumptions underlying the alternative policy scenario are as follows:

- Congo is assumed to maintain a minimum level of official foreign reserves equal to 5 months of imports.
- In the face of declining oil revenues and continued large government spending, it is assumed that capital expenditures are financed by first running down deposits in China. Following this, the government increases external borrowing to sustain higher government spending than in the baseline. As a result, the debt-to-GDP ratio is projected to exceed 50 percent by 2021 and to top 66 percent by 2025. Under the alternative policy scenario, new loans would likely be less concessional from 2020 onwards. Interest rates would rise to 5 percent, on average, and the average grace period would shorten. Eventually, the rise in external debt would force the inevitable downward adjustment in government spending.
- The longer-term effect of postponing the fiscal consolidation is to run up unsustainable external liabilities, leaving no financial assets to provide a permanent income once oil reserves are exhausted.

CONCLUSION

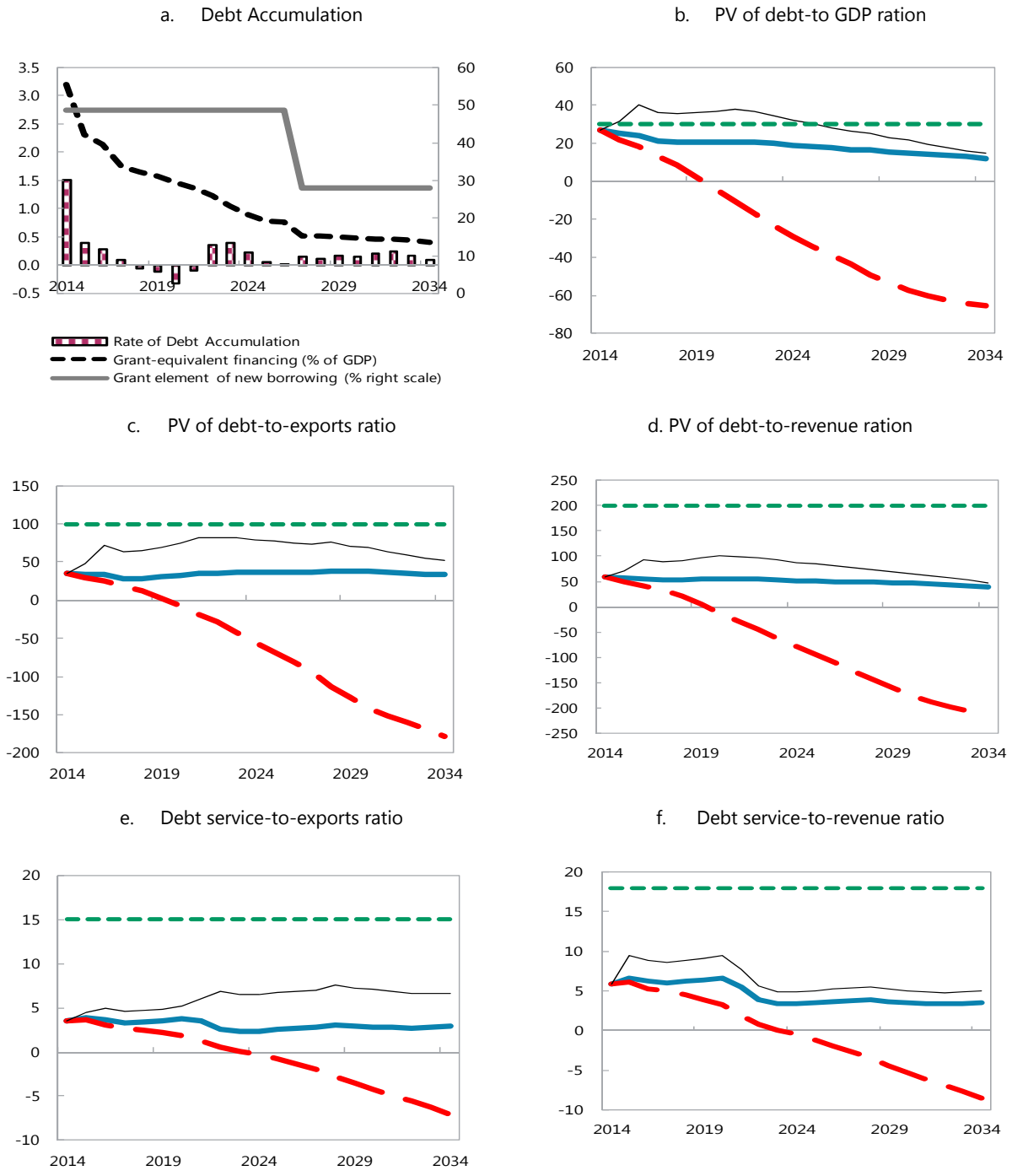
13. In the Staff's view, the Republic of Congo continues to face a low risk of debt distress, but with the potential of a negative revision in the future if external vulnerabilities deteriorate. All external debt indicators are well below the country-specific indicative thresholds under the baseline "strong policy" scenario but one of these thresholds is temporarily breached under the various standard stress tests. Nevertheless, Congo is unlikely to resort to extensive external borrowing in the near future given the comfortable level of reserves. It is therefore important for the government to anchor macroeconomic policies by a stability-oriented medium-term fiscal framework that targets a gradual reduction of the non-oil primary fiscal deficit and to save some oil revenue to prepare for the post-oil era.[18] The pattern seen in the last 3 years with respect to external borrowing underscores the constrained capacity to finance large infrastructure projects. Since we expect this trend to continue, new borrowing contracts should reflect similar concessional terms as the recent ones in order to maintain debt sustainability. In a "alternative-policy" scenario, which illustrates the risks of persistent delays in putting the non-oil primary deficit on a medium-term consolidation path, vulnerabilities would increase and the Republic of Congo would need to be reclassified to being at a "high risk of debt distress."

14. Improving competitiveness and promoting economic diversification are key to increasing resilience to exogenous shocks. As the DSA indicates, Congo's debt ratios appear to be most sensitive to swings in exports. Given the high concentration and vulnerability of the economy to downward movements in oil prices, broadening of the economic base by enhancing the development of the non-oil sector would reduce the volatility of exports and would strengthen the debt service capacity of the Congolese economy. In this regard, efforts toward improving the business climate with assistance from the World Bank should help strengthen competitiveness and boost growth. In the same vein, in addressing Congo's weak physical and human capital, priority should be given to making electricity supply more reliable and competitive, and to enhancing the quality of transportation services and of the labor force.

15. Focusing on policies to strengthen public debt management and institutions could enhance the Congolese authorities' capacity to manage debt and fiscal policies. In the DSA framework, this would raise the Congolese economies' debt-carrying capacity, by raising the country's CPIA rating. Moreover the development and integration of the regional debt market within the CEMAC area would broaden the menu of options to manage the Republic of Congo's financing needs.

[18] Although gross official reserves declined in the past two years, the government accumulated savings overseas.

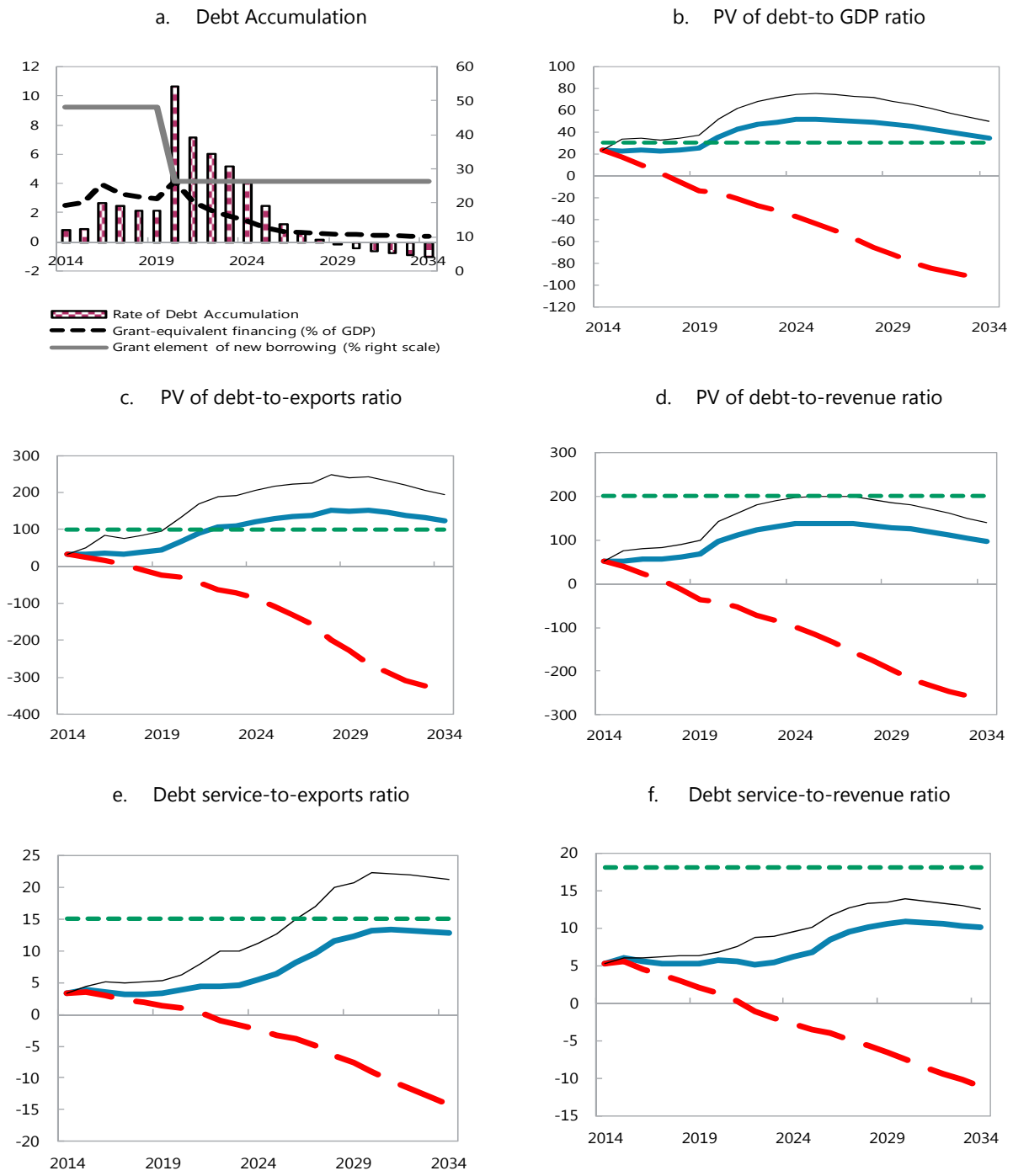
Figure 2a. Republic of Congo: Indicators of Public and Publicly Guaranteed External Debt under Baseline Scenario, 2014–34 [1/]



Sources: Country authorities; and staff estimates and projections.

[1/] The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a Exports shock; in c. to a Exports shock; in d. to a Exports shock; in e. to a Exports shock and in figure f. to a Exports shock

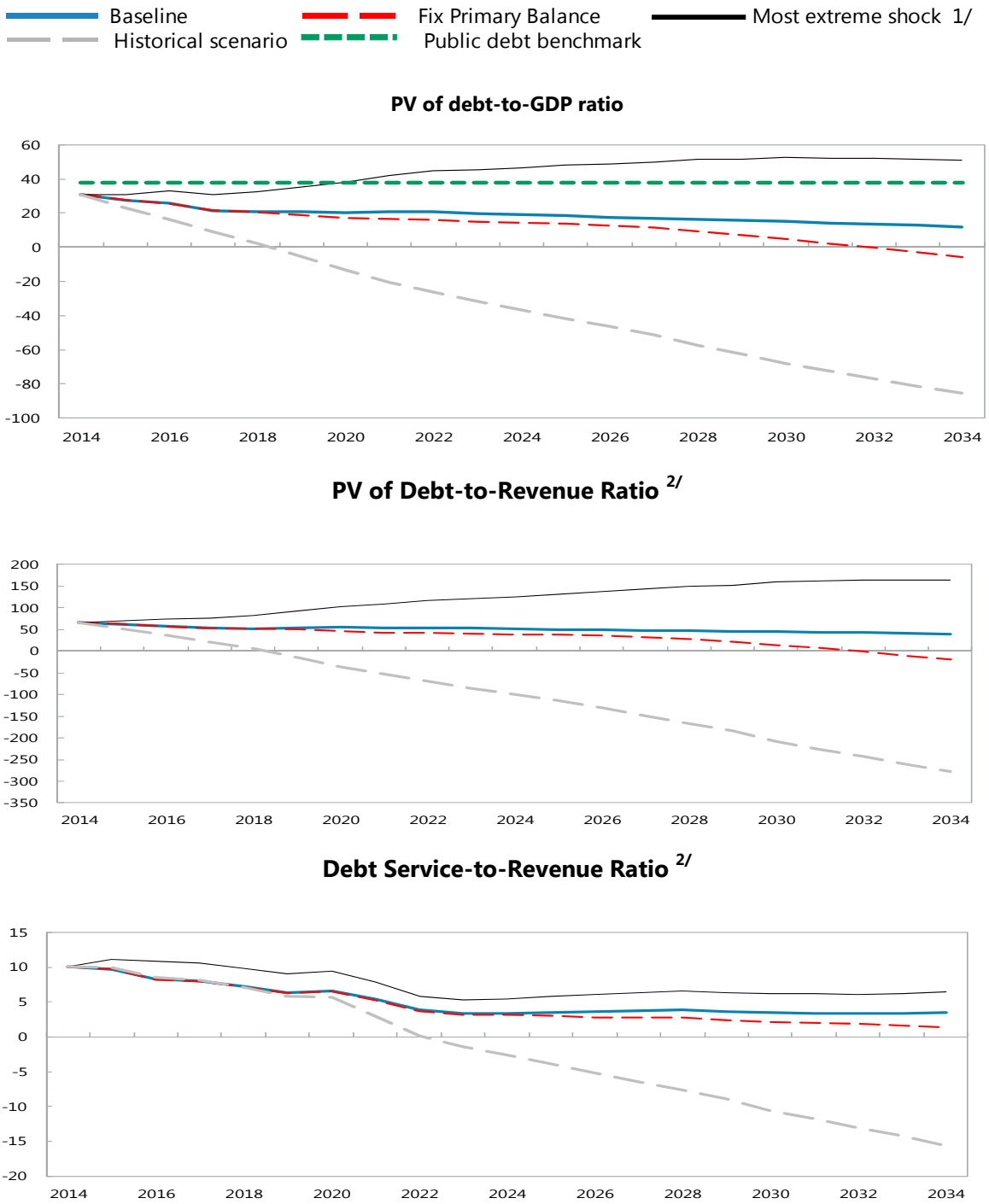
Figure 2b. Republic of Congo: Indicators of Public and Publicly Guaranteed External Debt under Alternative Policy Scenario, 2014–34 [1/]



Sources: Country authorities; and staff estimates and projections.

[1/] The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a Exports shock; in c. to a Exports shock; in d. to a Exports shock;

Figure 3. Republic of Congo: Indicators of Public Debt Under Alternative Shocks, Baseline Scenario, 2014–34



Sources: Congolese authorities; IMF and WB staff calculations and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.

Table 3. Republic of Congo: External Debt Sustainability Framework, Baseline Scenario, 2011–34

(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average [6/]	Standard Deviation [6/]	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-2019 Average	2024	2034	2020-203 Average
External debt (nominal) [1/]	**23.5**	**25.7**	**32.0**			**33.9**	**33.1**	**32.6**	**29.1**	**28.9**	**29.5**		**27.9**	**15.3**	
of which: public and publicly guaranteed (PPG)	23.5	25.7	32.0			33.9	33.1	32.6	29.1	28.9	29.5		27.9	15.3	
Change in external debt	3.2	2.2	6.4			1.8	-0.8	-0.5	-3.5	-0.2	0.6		-1.3	-1.2	
Identified net debt-creating flows	-30.5	-13.8	-15.1			-18.3	-16.5	-18.4	-16.6	-13.9	-9.7		-5.1	-2.2	
Non-interest current account deficit	**-6.1**	**1.1**	**3.2**	**-0.5**	**4.9**	**3.1**	**3.5**	**1.4**	**-2.3**	**-1.4**	**1.6**		**2.2**	**1.8**	2.2
Deficit in balance of goods and services	-27.2	-25.4	-17.5			-14.7	-14.2	-15.9	-21.8	-20.5	-16.2		-7.6	-1.4	
Exports	84.0	81.1	78.9			77.4	75.5	73.5	74.1	71.7	67.8		52.3	36.6	
Imports	56.8	55.7	61.4			62.7	61.3	57.5	52.3	51.1	51.6		44.7	35.2	
Net current transfers (negative = inflow)	2.0	2.8	2.9	1.7	1.1	2.7	3.0	3.0	2.8	3.0	3.1		3.1	2.1	2.8
of which: official	-0.4	-0.1	-0.3			-0.5	-0.2	-0.3	-0.2	-0.3	-0.3		-0.2	-0.1	
Other current account flows (negative = net inflow)	19.1	23.6	17.8			15.1	14.7	14.3	16.6	16.2	14.7		6.7	1.1	
Net FDI (negative = inflow)	**-21.1**	**-16.4**	**-18.9**	**-17.5**	**7.9**	**-19.8**	**-18.0**	**-17.8**	**-11.9**	**-11.8**	**-11.0**		**-6.3**	**-3.4**	-5.4
Endogenous debt dynamics [2/]	**-3.2**	**1.5**	**0.6**			**-1.6**	**-2.0**	**-2.1**	**-2.3**	**-0.8**	**-0.3**		**-0.9**	**-0.6**	
Contribution from nominal interest rate	0.2	0.2	0.2			0.2	0.4	0.2	0.2	0.2	0.2		0.1	0.1	
Contribution from real GDP growth	-0.6	-0.9	-0.9			-1.8	-2.3	-2.3	-2.5	-1.0	-0.5		-1.1	-0.7	
Contribution from price and exchange rate changes	-2.8	2.2	1.2			…	…	…	…	…	…		…	…	
Residual (3-4) [3/]	**33.7**	**16.0**	**21.5**			**20.2**	**15.7**	**17.9**	**13.1**	**13.7**	**10.3**		**3.8**	**1.0**	
of which: exceptional financing	0.0	0.4	-0.5			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt [4/]	…	…	26.6			27.0	25.4	24.3	21.2	20.6	20.6		19.0	12.1	
In percent of exports	…	…	33.7			34.8	33.6	33.1	28.6	28.7	30.3		36.3	33.1	
PV of PPG external debt	**…**	**…**	**26.6**			**27.0**	**25.4**	**24.3**	**21.2**	**20.6**	**20.6**		**19.0**	**12.1**	
In percent of exports	**…**	**…**	**33.7**			**34.8**	**33.6**	**33.1**	**28.6**	**28.7**	**30.3**		**36.3**	**33.1**	
In percent of government revenues	**…**	**…**	**57.2**			**58.6**	**57.5**	**55.9**	**52.7**	**52.8**	**54.7**		**51.9**	**39.6**	
Debt service-to-exports ratio (in percent)	**0.2**	**0.2**	**0.3**			**3.5**	**3.9**	**3.7**	**3.3**	**3.4**	**3.5**		**2.4**	**2.9**	
PPG debt service-to-exports ratio (in percent)	**0.2**	**0.2**	**0.3**			**3.5**	**3.9**	**3.7**	**3.3**	**3.4**	**3.5**		**2.4**	**2.9**	
PPG debt service-to-revenue ratio (in percent)	**0.4**	**0.4**	**0.5**			**5.9**	**6.6**	**6.2**	**6.1**	**6.2**	**6.4**		**3.4**	**3.5**	
Total gross financing need (Billions of U.S. dollars)	-3.9	-2.1	-2.1			-2.0	-1.8	-2.2	-2.2	-2.1	-1.3		-0.6	-0.2	
Non-interest current account deficit that stabilizes debt ratio	-9.3	-1.2	-3.2			1.2	4.3	1.9	1.1	-1.2	1.0		3.5	3.0	
Key macroeconomic assumptions															
Real GDP growth (in percent)	3.4	3.8	3.3	4.8	3.0	6.0	7.5	7.3	9.0	3.5	1.9	5.8	3.9	4.7	3.7
GDP deflator in US dollar terms (change in percent)	16.0	-8.7	-4.6	10.7	18.2	0.6	0.6	-1.8	5.6	-0.6	-2.3	0.4	1.3	2.2	0.7
Effective interest rate (percent) [5/]	0.9	0.8	0.9	2.5	1.4	0.6	1.2	0.7	0.8	0.7	0.7	0.8	0.5	0.6	0.5
Growth of exports of G&S (US dollar terms, in percent)	21.6	-8.5	-4.1	15.7	24.4	4.6	5.4	2.5	16.0	-0.5	-5.8	3.7	1.0	3.3	0.3
Growth of imports of G&S (US dollar terms, in percent)	18.6	-7.0	8.6	17.5	21.5	8.9	5.7	-1.2	4.6	0.6	0.5	3.2	2.0	4.1	1.8
Grant element of new public sector borrowing (in percent)	…	…	…	…	…	48.7	48.7	48.7	48.7	48.7	48.7	48.7	48.7	28.1	37.7
Government revenues (excluding grants, in percent of GDP)	42.0	42.5	46.5			46.1	44.2	43.5	40.3	39.0	37.6		36.6	30.6	34.5
Aid flows (in Billions of US dollars) [7/]	0.1	0.0	0.1			0.1	0.0	0.1	0.1	0.1	0.1		0.1	0.1	
of which: Grants	0.1	0.0	0.1			0.1	0.0	0.1	0.1	0.1	0.1		0.1	0.1	
of which: Concessional loans	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Grant-equivalent financing (in percent of GDP) [8/]	…	…	…			3.2	2.3	2.1	1.8	1.7	1.6		0.9	0.4	0.8
Grant-equivalent financing (in percent of external financing) [8/]	…	…	…			53.4	52.3	53.1	53.1	54.0	54.4		58.0	39.5	48.3
Memorandum items:															
Nominal GDP (Billions of US dollars)	14.4	13.7	13.5			14.4	15.5	16.4	18.8	19.4	19.3		21.5	36.9	
Nominal dollar GDP growth	20.0	-5.2	-1.5			6.6	8.1	5.4	15.1	2.8	-0.4	6.3	5.2	7.0	4.5
PV of PPG external debt (in Billions of US dollars)			3.7			3.9	4.0	4.0	4.0	4.0	4.0		4.1	4.5	
(PVt-PVt-1)/GDPt-1 (in percent)						1.5	0.4	0.3	0.1	0.0	-0.1	0.4	0.2	0.1	0.1
Gross workers' remittances (Billions of US dollars)	…	…	…			…	…	…	…	…	…		…	…	
PV of PPG external debt (in percent of GDP + remittances)	…	…	26.6			27.0	25.4	24.3	21.2	20.6	20.6		19.0	12.1	
PV of PPG external debt (in percent of exports + remittances)	…	…	33.7			34.8	33.6	33.1	28.6	28.7	30.3		36.3	33.1	
Debt service of PPG external debt (in percent of exports + remittances)	…	…	0.3			3.5	3.9	3.7	3.3	3.4	3.5		2.4	2.9	

Sources: Congolese authorities; IMF and WB staff calculations and projections.

[1/] Includes both public and private sector external debt.

[2/] Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

[3/] Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate ch

[4/] Assumes that PV of private sector debt is equivalent to its face value.

[5/] Current-year interest payments divided by previous period debt stock.

[6/] Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

[7/] Defined as grants, concessional loans, and debt relief.

[8/] Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 4. Republic of Congo: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, Baseline Scenario, 2014–34

(In percent)

	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
PV of debt-to GDP ratio								
Baseline	27	26	25	21	21	21	**19**	12
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	27	23	21	16	12	6	**-7**	-8
A2. New public sector loans on less favorable terms in 2014-2034 2	27	27	26	24	24	24	**25**	18
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	27	27	27	24	23	23	**21**	14
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	27	31	40	35	35	35	**31**	14
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	27	28	28	25	24	24	**22**	14
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	27	30	34	30	29	30	**27**	14
B5. Combination of B1-B4 using one-half standard deviation shocks	27	29	31	27	27	27	**24**	13
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	27	37	35	31	30	30	**28**	18
PV of debt-to-exports ratio								
Baseline	35	34	34	29	29	31	**37**	34
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	35	31	29	22	17	9	**-13**	-23
A2. New public sector loans on less favorable terms in 2014-2034 2	35	35	36	32	33	36	**48**	51
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	35	34	34	29	29	31	**37**	34
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	35	47	69	61	62	67	**77**	51
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	35	34	34	29	29	31	**37**	34
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	35	40	46	41	41	44	**51**	37
B5. Combination of B1-B4 using one-half standard deviation shocks	35	38	41	36	37	39	**46**	36
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	35	34	34	29	29	31	**37**	34
PV of debt-to-revenue ratio								
Baseline	58	58	57	54	54	56	**53**	40
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	58	53	49	41	31	17	**-18**	-27
A2. New public sector loans on less favorable terms in 2014-2034 2	58	60	61	59	61	65	**68**	60
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	58	62	63	60	60	62	**59**	45
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	58	71	91	88	90	95	**86**	47
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	58	63	65	62	62	64	**61**	46
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	58	69	78	75	76	80	**73**	44
B5. Combination of B1-B4 using one-half standard deviation shocks	58	67	72	69	69	73	**67**	44
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	58	83	82	77	77	80	**76**	57

Table 5. Republic of Congo: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, Baseline Scenario, 2014–34 *(continued*

(In percent)

	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
Debt service-to-exports ratio								
Baseline	3	4	4	3	3	4	**2**	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 [1/]	3	4	3	3	2	2	**0**	-3
A2. New public sector loans on less favorable terms in 2014-2034 [2/]	3	4	4	3	4	4	**3**	4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	3	4	4	3	3	4	**2**	3
B2. Export value growth at historical average minus one standard deviation in 2015-2016 [3/]	3	4	5	5	5	5	**7**	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	3	4	4	3	3	4	**2**	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 [4/]	3	4	4	3	4	4	**4**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	3	4	4	3	4	4	**3**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 [5/]	3	4	4	3	3	4	**2**	3
Debt service-to-revenue ratio								
Baseline	6	7	6	6	6	6	**3**	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 [1/]	6	6	5	5	4	4	**0**	-3
A2. New public sector loans on less favorable terms in 2014-2034 [2/]	6	7	6	6	7	7	**4**	5
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	6	7	7	7	7	7	**4**	4
B2. Export value growth at historical average minus one standard deviation in 2015-2016 [3/]	6	7	6	6	7	7	**7**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	6	7	7	7	7	7	**4**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 [4/]	6	7	6	6	7	7	**6**	5
B5. Combination of B1-B4 using one-half standard deviation shocks	6	7	7	6	7	7	**5**	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 [5/]	6	9	9	9	9	9	**5**	5
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) [6/]	40	40	40	40	40	40	**40**	40

Sources: Congolese authorities; IMF and WB staff calculations and projections.

[1/] Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

[2/] Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

[3/] Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly a an offsetting adjustment in import levels).

[4/] Includes official and private transfers and FDI.

[5/] Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

[6/] Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 6. Republic of Congo: Public Sector Debt Sustainability Framework, Baseline Scenario, 2011–34

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2011	2012	2013	Average [5/]	Standard Deviation [5/]	2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
Public sector debt [1/]	33.1	34.1	38.2			37.9	35.4	33.9	29.5	28.9	29.5		27.9	15.3	
of which: foreign-currency denominated	23.5	25.7	32.0			33.9	33.1	32.6	29.1	28.9	29.5		27.9	15.3	
Net public debt	33.1	34.1	38.2			37.9	35.4	33.9	29.5	28.9	29.5		27.9	15.3	
Change in public sector debt	10.2	1.1	4.1			-0.3	-2.4	-1.5	-4.4	-0.6	0.6		-1.3	-1.2	
Identified debt-creating flows	-19.3	-6.0	-5.5			-6.7	-7.6	-6.7	-10.3	-5.8	-2.7		-6.6	-2.3	
Primary deficit	-16.6	-6.6	-8.8	-14.3	6.8	-5.5	-5.3	-5.1	-6.0	-5.2	-3.1	-5.0	-5.2	-1.4	-3.4
Revenue and grants	42.5	42.6	46.9			46.6	44.5	43.9	40.6	39.3	37.9		36.9	30.8	
of which: grants	0.5	0.1	0.4			0.6	0.3	0.3	0.3	0.3	0.3		0.3	0.2	
Primary (noninterest) expenditure	25.9	36.0	38.1			41.1	39.2	38.8	34.5	34.1	34.9		31.7	29.4	
Automatic debt dynamics	-2.7	-0.5	0.8			-1.2	-2.3	-1.6	-4.3	-0.6	0.3		-1.3	-1.0	
Contribution from interest rate/growth differential	-1.2	-1.3	-0.5			-2.3	-2.8	-2.7	-3.2	-1.3	-0.9		-1.5	-1.0	
of which: contribution from average real interest rate	-0.4	-0.1	0.6			-0.1	-0.1	-0.3	-0.4	-0.3	-0.4		-0.4	-0.2	
of which: contribution from real GDP growth	-0.8	-1.2	-1.1			-2.2	-2.6	-2.4	-2.8	-1.0	-0.5		-1.1	-0.7	
Contribution from real exchange rate depreciation	-1.5	0.8	1.3			1.0	0.5	1.1	-1.1	0.7	1.2		...	...	
Other identified debt-creating flows	0.0	1.1	2.5			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	1.1	2.5			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	29.5	7.1	9.6			6.4	5.2	5.2	5.9	5.2	3.3		5.3	1.2	
Other Sustainability Indicators															
PV of public sector debt	...	...	32.8			31.0	27.8	25.7	21.6	20.6	20.6		19.0	12.1	
of which: foreign-currency denominated	...	...	26.6			27.0	25.4	24.3	21.2	20.6	20.6		19.0	12.1	
of which: external	...	...	26.6			27.0	25.4	24.3	21.2	20.6	20.6		19.0	12.1	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need [2/]	-16.5	-5.5	-5.8			-0.8	-1.0	-1.5	-2.8	-2.4	-0.7		-4.0	-0.3	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	69.9			66.5	62.4	58.6	53.3	52.3	54.3		51.5	39.4	
PV of public sector debt-to-revenue ratio (in percent)	...	...	70.5			67.3	62.8	59.0	53.7	52.8	54.7		51.9	39.6	
of which: external [3/]	...	...	57.2			58.6	57.5	55.9	52.7	52.8	54.7		51.9	39.6	
Debt service-to-revenue and grants ratio (in percent) [4/]	0.4	2.6	6.4			10.1	9.7	8.2	8.0	7.2	6.3		3.4	3.5	
Debt service-to-revenue ratio (in percent) [4/]	0.4	2.6	6.4			10.2	9.8	8.3	8.0	7.2	6.4		3.4	3.5	
Primary deficit that stabilizes the debt-to-GDP ratio	-26.8	-7.7	-12.9			-5.1	-2.9	-3.6	-1.6	-4.6	-3.6		-3.9	-0.2	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	3.4	3.8	3.3	4.8	3.0	6.0	7.5	7.3	9.0	3.5	1.9	5.8	3.9	4.7	3.7
Average nominal interest rate on forex debt (in percent)	0.9	0.8	0.9	2.5	1.4	0.6	1.2	0.7	0.8	0.7	0.7	0.8	0.5	0.6	0.5
Average real interest rate on domestic debt (in percent)	...	...	8.8	...	...	...	...	...	...	...	...		...	...	
Real exchange rate depreciation (in percent, + indicates depreciation)	-7.6	3.5	5.3	-5.4	12.0	3.4	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	10.7	-1.2	-7.7	8.5	15.0	-2.9	-1.2	-2.7	4.2	-2.3	-3.9	-1.5	1.3	2.2	0.7
Growth of real primary spending (deflated by GDP deflator, in percent)	31.0	44.2	9.4	8.5	15.9	14.4	2.4	6.1	-3.0	2.2	4.2	4.4	0.8	3.0	2.6
Grant element of new external borrowing (in percent)	...	...	...	...	...	48.7	48.7	48.7	48.7	48.7	48.7	48.7	48.7	28.1	...

Sources: Congolese authorities; IMF and WB staff calculations and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 7. Republic of Congo: Sensitivity Analysis for Key Indicators of Public Debt, Baseline Scenario 2014–34

		Projections						
	2014	2015	2016	2017	2018	2019	2024	2034
PV of Debt-to-GDP Ratio								
Baseline	31	28	26	22	21	21	19	12
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	31	23	16	9	2	-5	-37	-86
A2. Primary balance is unchanged from 2014	31	28	25	22	20	19	14	-6
A3. Permanently lower GDP growth [1/]	31	28	26	23	23	24	29	39
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-201	31	31	33	31	32	35	47	51
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	31	26	23	19	18	18	16	10
B3. Combination of B1-B2 using one half standard deviation shocks	31	26	21	19	20	22	29	34
B4. One-time 30 percent real depreciation in 2015	31	38	34	28	27	26	23	15
B5. 10 percent of GDP increase in other debt-creating flows in 2015	31	34	32	27	26	26	25	16
PV of Debt-to-Revenue Ratio [2/]								
Baseline	66	62	59	53	52	54	51	39
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	66	52	37	22	6	-14	-100	-278
A2. Primary balance is unchanged from 2014	66	62	58	53	52	50	38	-18
A3. Permanently lower GDP growth [1/]	66	63	60	56	57	62	77	126
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-201	66	69	75	76	82	93	126	165
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	66	59	52	47	46	47	44	34
B3. Combination of B1-B2 using one half standard deviation shocks	66	58	48	48	51	57	80	110
B4. One-time 30 percent real depreciation in 2015	66	85	78	70	68	69	63	49
B5. 10 percent of GDP increase in other debt-creating flows in 2015	66	76	72	66	66	69	67	51
Debt Service-to-Revenue Ratio [2/]								
Baseline	10	10	8	8	7	6	3	3
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	10	10	8	8	7	6	-3	-16
A2. Primary balance is unchanged from 2014	10	10	8	8	7	6	3	1
A3. Permanently lower GDP growth [1/]	10	10	8	8	7	7	4	8
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-201	10	10	9	9	8	7	7	12
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	10	10	8	8	7	6	3	3
B3. Combination of B1-B2 using one half standard deviation shocks	10	10	9	8	8	7	4	7
B4. One-time 30 percent real depreciation in 2015	10	11	11	11	10	9	5	6
B5. 10 percent of GDP increase in other debt-creating flows in 2015	10	10	8	8	7	6	5	5

Sources: Congolese authorities; IMF and WB staff calculations and projections.

[1/] Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

[2/] Revenues are defined inclusive of grants.

Table 8. Republic of Congo: Comparison of Baseline and Alternative Policy Scenarios

	Averages					
	2014–19	2020–26	2027–34	2014–19	2020–26	2027–34
	Baseline			Alternative Policy		
Production and Prices	(Annual percentage change)					
Real GDP	5.8	3.3	4.1	6.1	2.4	1.8
Non-oil real GDP	6.6	5.4	4.9	6.9	4.5	2.5
Consumer prices (end of period)	2.8	3.0	2.1	3.6	2.9	1.6
External Sector	(Percent of GDP)					
Current account balance	-1.2	-1.9	-2.7	-7.1	-6.4	-4.1
Gross official foreign reserves	46.5	53.8	38.9	39.8	25.6	21.9
Gross official foreign reserves (months of imports)	10.1	13.9	12.0	7.7	6.5	6.5
External public debt	34.5	34.1	22.7	37.3	59.8	65.8
Fiscal Aggregates	(Percentage of non-oil GDP)					
Nonoil revenue (inc. grants and investment income)	30.6	31.3	30.9	30.5	31.2	31.4
Total expenditure	75.8	45.1	32.9	93.4	54.9	39.0
Overall balance (deficit -, commitment basis) [1/]	9.8	6.2	2.4	-9.1	-4.9	-2.4
Basic primary fiscal balance (- = deficit) [2/]	13.6	6.4	2.9	-5.3	-3.4	0.6
Basic non-oil primary balance (- = deficit) [3/]	-37.6	-11.2	-0.9	-52.5	-12.9	0.3
Basic non-oil primary balance (cash basis)	-39.6	-11.2	-0.9	-54.4	-12.9	0.3
Memorandum items:	(Billions of CFA Francs)					
Nominal GDP	7,936	9,163	13,203	8,074	9,436	10,992
Government expenditure and net lending	2,943	2,975	4,024	3,838	3,791	3,923
Government expenditure (growth in percent)	2.8	0.9	5.4	10.2	-2.4	1.5

Sources: Congolese authorities; and Fund staff estimates and projections.

[1/] Including grants.

[2/] Revenue (excluding investment income and grants) minus total expenditures (excluding interest payments and foreign-financed investment.

[3/] Basic primary balance minus oil revenue.